

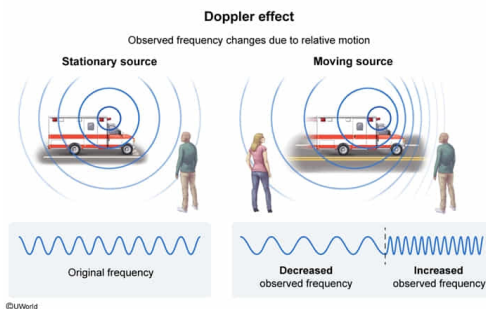
An observer perceives a sound to be lower in frequency when the sound source is moving away from the observer. This shift in the frequency of the perceived sound occurs because:

- ☐ A. the sound waveform velocity decreases as the distance between source and observer increases. [26%]
- ☐ B. sound waveforms are reflected off environmental objects nearby. [1%]
- ✓ ☒ C. successive sound waveforms are emitted further away from the observer. [35%]
- ☐ D. the sound wave intensity decreases as the distance between source and observer increases. [37%]

Correct

35% answered correctly

Explanation:



The **Doppler effect** describes a waveform phenomenon in which the frequency of the waveforms perceived by an observer is greater or less than the real, native frequency of waveforms emitted from the source. The Doppler effect may be observed with both longitudinal (eg, sound) and transverse (eg, light) waveforms because both exhibit the same relationship among frequency (f), native velocity (v), and wavelength (λ):

$$f = \frac{v}{\lambda}$$

Doppler shifts occur when the velocity of the waveform source (v_s) or the velocity of an observer (v_o) leads to a perceived change in the velocity of the emitted waveforms relative to the observer. A perceived change in v resulting from either cause produces a simultaneous apparent shift in the observed frequency (f_o):

$$f_o = \frac{v \pm v_o}{\lambda} = \frac{v \pm v_o}{\frac{v}{f \pm \frac{v_s}{\lambda}}}$$

In other words, the motion of the observer or waveform source relative to each other leads to a shift in only the **apparent, perceived velocity** of the sound waveform as successive waveform **crests and troughs** are perceived closer together or farther apart. For example, when the source or observer is moving away from the other, the distance between successive waveforms increases and the perceived waveform velocity decreases.

(Choice A) Doppler shifts describe only the *perceived velocity* of emitted waveforms. No amount of motion by either the source or the observer can modify the native velocity of the emitted waveform, because the speed of a waveform is primarily influenced by its type and by the medium through which it travels.

(Choice B) The scattering of sound that occurs with the propagation of sound through open environments is associated with reduced waveform intensity, but this is not responsible for the frequency shift associated with the Doppler effect.

(Choice D) The **decay in soundwave intensity** that occurs as the distance increases between the source and the observer is unrelated to the motion of the source and the observer; it cannot explain the frequency shifts associated with the Doppler effect.

Educational objective:

The Doppler effect describes apparent changes in the velocity of waveforms that lead to changes in the observed frequency of those waveforms, which are known as Doppler shifts; however, the actual velocity of waveforms does not change when Doppler shifts occur.

Time spent:0 sec

Subject:

Physics

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System:

Light and Sound